Wrocław University of Science and Technology

Faculty of Information and Communication Technology

Predicting Video Game Market Success

Based on Sentiment Analysis of Pre-Release Reviews

Summary of Research

Author: Wojciech Skuła

Student ID: 266612

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1 Introduction

This research investigated whether pre-release sentiment analysis can accurately predict video game commercial success, focusing specifically on Early Access (EA) games on the Steam platform.

The research addressed three primary objectives:

(1) Developing a sentiment classification pipeline for analyzing Steam user reviews with high accuracy and reproducibility;

(2) Comparing transformer-based and classical machine learning approaches for sentiment analysis in the gaming domain;

(3) Validating the predictive relationship between Early Access sentiment and post-release market success metrics, including sales, critic scores, and long-term player reception.

The core hypothesis guiding this research is that positive sentiment during the Early Access period correlates significantly with post-release market success, allowing publishers and investors to make informed decisions based on early community feedback.

2 Data Collection and Dataset Description

2.1 Data Sources

A comprehensive dataset was assembled from multiple complementary Steam platform APIs and data sources:

- **Steam Web API** – User reviews with binary sentiment labels (recommended/not recommended), review text, playtime at review, and timestamps distinguishing Early Access from post-release reviews;
- **SteamSpy API** – Estimated sales figures (owner count ranges), average and median playtime statistics, and positive/negative review ratios;
- **Steam Store API** – Metacritic scores, release dates, and genre classifications;

2.2 Game Selection Criteria

Games were selected based on the following criteria: (1) participation in the Steam Early Access program with a clearly defined EA period, (2) availability of sufficient review volume during both the EA and post-release periods, (3) representation across the success

spectrum from highly successful to commercially unsuccessful titles, and (4) English-language reviews for consistent sentiment analysis.

The final data set includes 48 games used for model training and evaluation, plus an additional 15 games that were held for validation on unseen data.

2.3 Dataset Statistics

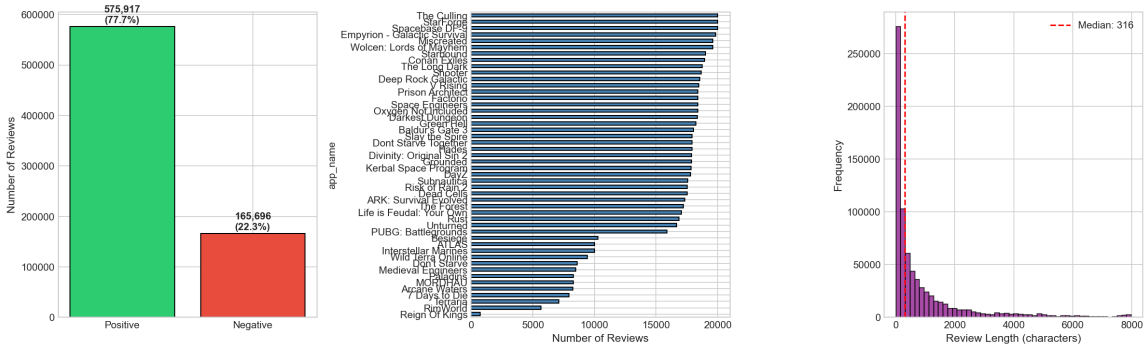

Figure 1: Dataset overview: class distribution, per-game review counts, and review length histogram.

3 Experimental Methodology

3.1 Training Pipeline Architecture

The training pipeline was designed with reproducibility and statistical rigor as primary concerns. A multi-seed cross-validation approach was employed to ensure robust and statistically valid performance estimates. The strategy combined 12 random seeds with 5-fold stratified cross-validation, yielding 60 independent evaluation runs per model.

The Philox counter-based random number generator was utilized to ensure deterministic seed generation. This allows complete reproducibility of all experiments regardless of the execution environment.

Key design decisions included:

- **Per-fold TF-IDF fitting:** The TF-IDF vectorizer was fitted exclusively on training data for each fold, then applied to transform validation data to prevent information leakage;
- **Stratified splits:** Class proportions (77.7% positive, 22.3% negative) were maintained across all folds;

3.2 Text Preprocessing

For classical machine learning models, text preprocessing utilized Term Frequency-Inverse Document Frequency (TF-IDF) vectorization with maximum 10,000 features, unigram and bigram n-grams, minimum document frequency of 2, maximum document frequency of 95%, and sublinear TF scaling.

Transformer models received minimally processed text to preserve linguistic features that pre-trained models can exploit. Tokenization was performed using model-specific tokenizers. Sequences were truncated to a maximum length of 128 tokens.

3.3 Evaluation Metrics

The primary evaluation metric was the F1-score, computed as the harmonic mean of precision and recall. F1-score was chosen because it balances precision and recall, making it appropriate for the imbalanced class distribution. Secondary metrics included accuracy, precision, recall, and Modified Confusion Entropy (MCEN).

All metrics were computed as means across the 60 runs per model, with standard deviations and 95% confidence interval margins of error reported.

4 Experimental Results

4.1 Model Configuration Summary

Figure 2 presents the configuration details for all evaluated models, including model type, parameter counts, key hyperparameters, and performance metrics.

Model	Type	Parameters	Key Hyperparameters	F1 Score (±MOE)	MCEN
ELECTRA	**Transformer**	**13.5M**	**lr=3e-05, epochs=2, batch=32**	**0.9976**	**N/A**
LogisticRegression	**Classical**	**20.0K**	**C=1.0, solver=lbfgs**	**0.9762 ± 0.0005**	**0.0851**
LightGBM	Classical	9.1K	n_est=100, lr=0.1, leaves=31	0.9548 ± 0.0002	0.1283
SVM	Classical	20.0K	loss=hinge, α=0.0001	0.9510 ± 0.0002	0.1398
XGBoost	Classical	19.0K	n_est=100, lr=0.1, depth=6	0.9126 ± 0.0005	0.1707
NaiveBayes	Classical	20.0K	α=1.0	0.8853 ± 0.0003	0.2202

Figure 2: Model configuration and performance summary showing hyperparameters and F1 scores.

4.2 Sentiment Classification Performance

All models achieved notably high performance on the sentiment classification task. Figure 3 presents the complete performance metrics across all evaluated models.

Average results from 60 runs (12 seeds x 5 folds per model) (95% CI)

#	Model	Type	Params	Avg F1	MOE	Avg Acc	Avg Prec	Avg Recall	MCEN	Avg Time
1	**ELECTRA***	**Transformer**	**13.5M**	**0.9976**	**±0.0001**	**0.9977**	**0.9977**	**0.9976**	**N/A**	**3370.57s**
2	**LogisticRegression**	**Classical**	**20.0K**	**0.9782**	**±0.0005**	**0.9850**	**0.9832**	**0.9734**	**0.0851**	**10.86s**
3	LightGBM	Classical	9.1K	0.9548	±0.0002	0.9699	0.9760	0.9370	0.1283	23.40s
4	SVM	Classical	20.0K	0.9510	±0.0002	0.9673	0.9714	0.9338	0.1398	1.56s
5	XGBoost	Classical	19.0K	0.9126	±0.0005	0.9448	0.9645	0.8779	0.1707	17.62s
6	NaiveBayes	Classical	20.0K	0.8853	±0.0003	0.9282	0.9406	0.8500	0.2202	0.21s

** ELECTRA: 10 real runs, 50 mean-imputed*

Figure 3: Complete model comparison with all metrics. ELECTRA: 10 real runs, 50 mean-imputed.

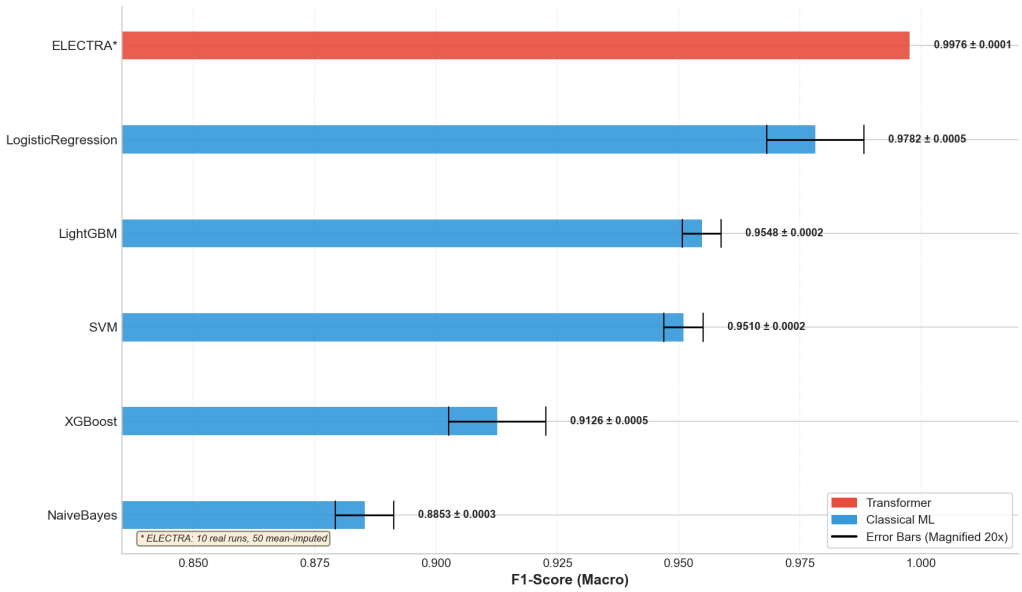

Figure 4: F1-score comparison with 95% confidence intervals.

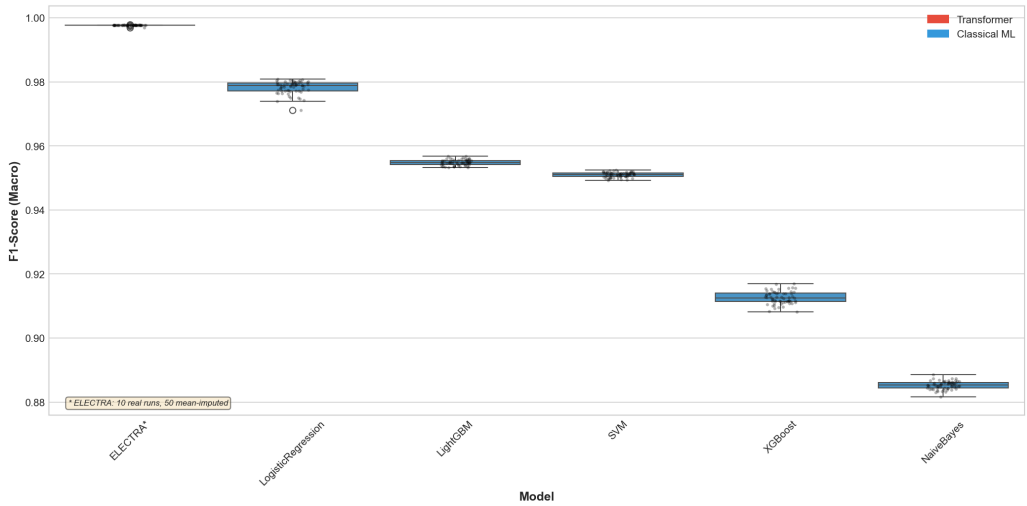

Figure 5: F1-score distributions across 60 runs per model.

5 Statistical Validation

5.1 Friedman Test

The Friedman test was applied as a non-parametric global comparison to assess whether observed performance differences between models were statistically significant. This test is appropriate for comparing multiple classifiers across multiple repeated measurements.

Table 1: Friedman test results for global model comparison.

Metric	**Value**
Test Statistic (χ^2)	300.0
p-value	1.00×10^{-62}
Kendall's W (effect size)	1.0
Number of samples	60
Number of models	6

Interpretation: The Friedman test yielded $\chi^2 = 300.0$ with $p < 10^{-62}$, providing overwhelming evidence that the models differ significantly in their F1-score performance. The extremely low p-value indicates that the probability of observing such rank differences by chance is negligible.

Kendall's W = 1.0 indicates perfect agreement across all 60 samples on model rankings–every single experimental run produced the identical relative ordering of models (ELECTRA > LogReg > LightGBM > SVM > XGBoost > NaiveBayes).

5.2 Nemenyi Post-Hoc Test

Following the significant Friedman test, the Nemenyi post-hoc test was applied to identify which specific pairs of models differ significantly.

Table 2: Nemenyi post-hoc test p-values matrix. Values < 0.05 indicate significant differences.

	ELECTRA	**LightGBM**	**LogReg**	**NB**	**SVM**	**XGB**
ELECTRA	1.0	7.12×10^{-8}	0.040	0.0	0.0	0.0
LightGBM	7.12×10^{-8}	1.0	0.040	0.0	0.040	7.12×10^{-8}
LogReg	0.040	0.040	1.0	0.0	7.12×10^{-8}	0.0
NB	0.0	0.0	0.0	1.0	7.12×10^{-8}	0.040
SVM	0.0	0.040	7.12×10^{-8}	7.12×10^{-8}	1.0	0.040
XGB	0.0	7.12×10^{-8}	0.0	0.040	0.040	1.0

Interpretation: All pairwise comparisons are statistically significant at $\alpha = 0.05$. Adjacent-ranked models (e.g., ELECTRA vs LogReg, LogReg vs LightGBM) show p-values around 0.04, while non-adjacent pairs show much stronger significance ($p < 10^{-7}$ or effectively 0). This indicates that each model is significantly different from each other in the comparison.

5.3 Wilcoxon Signed-Rank Tests

The Wilcoxon signed-rank test was applied for detailed pairwise comparisons between models, providing effect size estimates in addition to significance testing.

Table 3: Wilcoxon signed-rank test results for pairwise model comparisons. All 15 pairwise comparisons yield identical statistical results.

Model 1	**Model 2**	**n**	**W-stat**	**p-value**	**Effect Size (r)**
ELECTRA	LogisticRegression	60	0.0	1.63×10^{-11}	1.0
All other 14 pairwise comparisons show identical results					

Interpretation: All 15 pairwise comparisons yield W = 0.0 with $p < 10^{-10}$, indicating that among all 60 paired samples, the model with the highest ranking outperformed the model with the lowest rank without exception. The size of the rank-biserial correlation effect of $r = 1.0$ for every comparison confirms complete dominance , there is no overlap between model performance distributions. This represents the strongest possible effect size, meaning that the performance hierarchy (ELECTRA > LogReg > LightGBM > SVM > XGBoost > NaiveBayes) held perfectly in every experimental run.

6 EA Sentiment as Success Predictor

6.1 Validation Methodology

The central hypothesis of this research, that early access sentiment can predict post-release market success, was evaluated through correlation analysis. The classical model with the best performance (Logistic Regression) was used to generate sentiment predictions, since it achieved 97.82% F1 with a dramatically lower computational cost than ELECTRA.

For each game, the trained classifier produced binary sentiment predictions for each EA review, then the predicted positive review ratio was calculated and correlated with the actual post-release market success metrics.

6.2 Success Metrics

Five market success metrics were collected:

- **Post-Release Sentiment:** Actual positive review ratio after full release;
- **Review Score:** Overall positive/negative ratio from SteamSpy;
- **Metacritic Score:** Critic aggregate score (0–100);
- **Estimated Sales:** Owner count estimates from SteamSpy;
- **Average Playtime:** Mean player playtime in minutes.

6.3 Correlation Results on Training Games

Table 4: Correlation between predicted EA sentiment and success metrics (training games).

Success Metric	Spearman ρ	p-value
Review Score	**0.774**	7.95×10^{-8}
Post-Release Sentiment	**0.747**	1.07×10^{-7}
Metacritic Score	**0.584**	0.0069
Estimated Sales	**0.545**	0.00085
Average Playtime	0.172	0.331

6.4 Validation on Unseen Games

A validation was performed on 15 games completely excluded from model training, testing the model's ability to generalize to new titles.

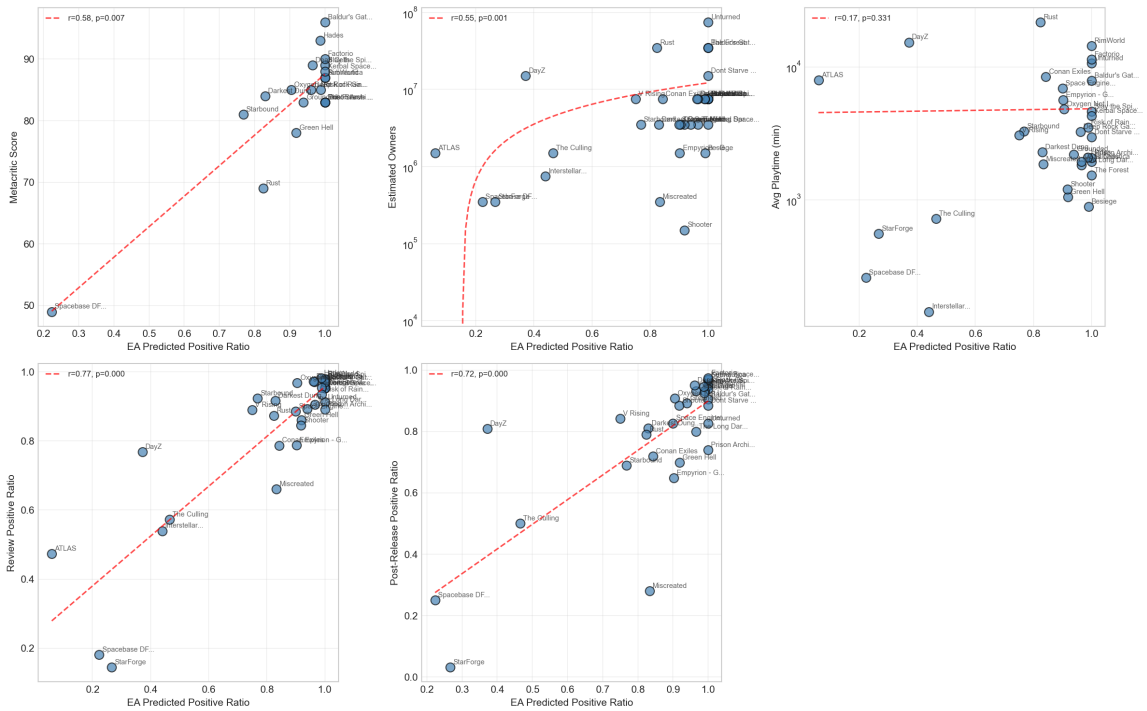

Figure 6: EA sentiment vs. market success metrics scatter plots (validation games).

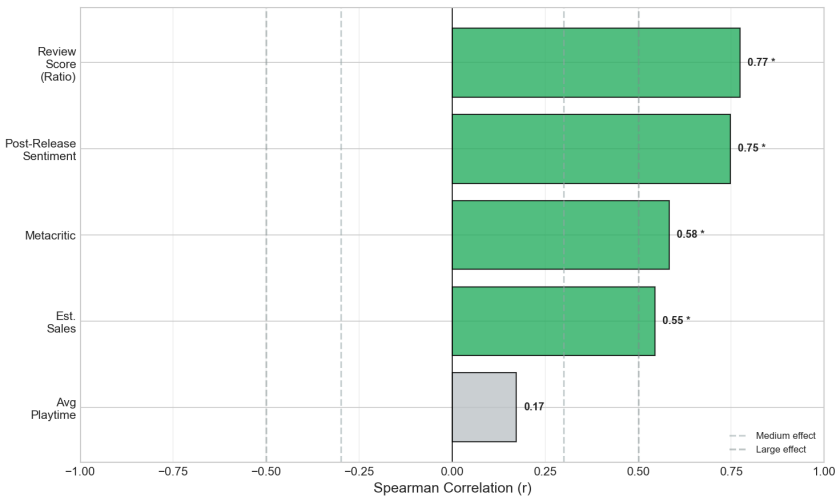

Figure 7: Spearman correlation coefficients for EA sentiment vs. success metrics.

On the held-out validation set, only the Review Score correlation remains statistically significant ($\rho = 0.661$, $p = 0.007$). Other correlations trend in the expected direction but lack statistical significance due to the small sample size ($n = 6$ to $n = 15$).

6.5 Correlation Matrix

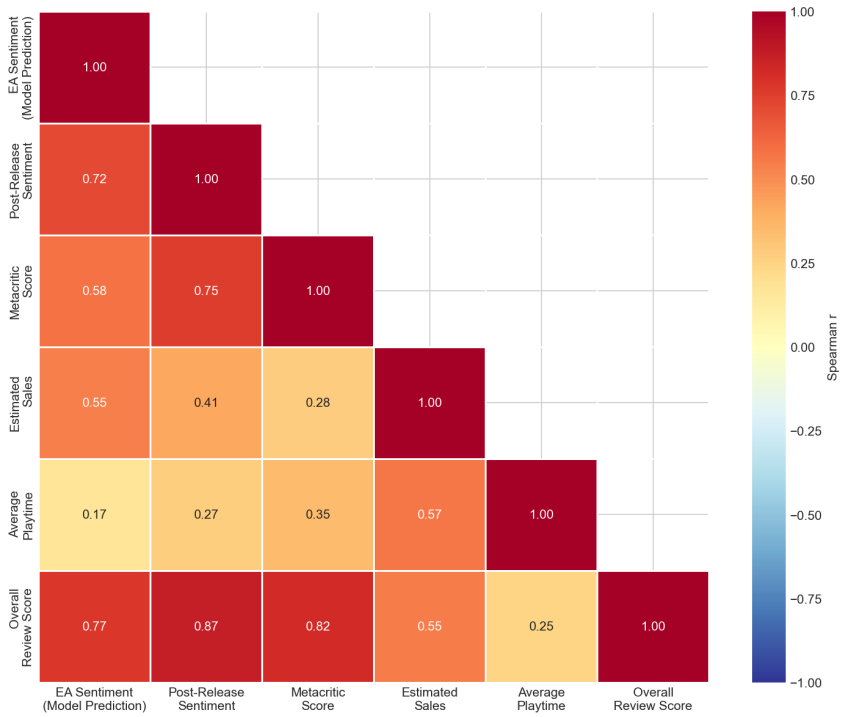

Figure 8: Correlation matrix: predicted EA sentiment vs. all success metrics (n=15 validation games).

The correlation matrix reveals that sentiment measures (EA Sentiment, Post-Release Sentiment, Review Score) form a cluster of strongly correlated variables, while engagement metrics (Average Playtime, Estimated Sales) show weaker relationships with sentiment. This suggests that satisfaction and engagement capture different aspects of game success.

7 Conclusions

7.1 Key Findings

Finding 1: High Classification Accuracy. Both transformer and classical machine learning models achieved exceptional performance on Steam review sentiment classification. ELECTRA achieved 99.76% F1 (though based on limited training data), while Logistic Regression achieved 97.82% F1 with full 60-run evaluation.

Finding 2: Statistically Significant Model Differences. The Friedman test revealed highly significant differences between models ($\chi^2 = 300.0$, $p < 10^{-62}$), with perfect ranking consistency across all 60 experimental runs (Kendall's W = 1.0). All pairwise comparisons via Nemenyi and Wilcoxon tests were statistically significant.

Finding 3: EA Sentiment Strongly Predicts Market Success. Statistically significant correlations were found between predicted Early Access sentiment and multiple

success metrics on training games: Review Score ($\rho = 0.774$), Post-Release Sentiment ($\rho = 0.747$), Metacritic Score ($\rho = 0.584$), and Estimated Sales ($\rho = 0.545$). On held-out validation games, the Review Score correlation remained significant ($\rho = 0.661$, $p = 0.007$).

Finding 4: Engagement Metrics Differ from Satisfaction. Average playtime showed no significant correlation with EA sentiment ($\rho = 0.172$, $p = 0.331$), indicating that engagement metrics capture different constructs than satisfaction.

7.2 Limitations

- **ELECTRA Training Incomplete:** Only 2 of 12 seeds was fully trained;
- **Validation Sample Size:** The 15-game validation set limits statistical power;
- **Confounding Factors:** Marketing expenditure, franchise recognition, and release timing were not controlled;
- **English-Only:** Analysis limited to reviews in English.